03 21826 Autumn 2025
LH The Life and Death of Stars

Lecture 2

(JCD 1.2-1.4)

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Previous lecture

- Knowledge of stellar physics crucial across all of astronomy
- Dynamical: $t_{\mathrm{dyn}} = \sqrt{\frac{2R^3}{GM}}$
- Thermal: $\tau_{th} = \frac{GM_*^2}{R_* L}$
- Nuclear: $t_{\mathrm{nuc}} = 7 \cdot 10^{-4} \frac{M_* c^2}{L_*}$
- $t_{\mathrm{dyn}} < \tau_{th} < t_{\mathrm{nuc}}$

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Outline

HR diagram

The life of a star

Stellar modeling

Summary

Outline

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Stellar modeling

Summary

Stellar luminosity and temperature

- Stars and their evolution are defined by observable properties [see more in Lecture 3]
- Stellar luminosity (lecture 1) can be measured through brightness and distance
- Effective temperature is defined through its relation with stellar luminosity and radius (σ is Stefan-Boltzmann constant):

$$L = 4\pi\sigma R^2 T_{\mathrm{eff}}^4$$

- Effective temperature is thus defined as the temperature the star would have if it would emit energy as black body radiation
- Effective temperature $\approx$ surface temperature $\neq$ internal temperature
- $T_{\mathrm{eff},\odot} = 5777\,\mathrm{K}$

The HR diagram

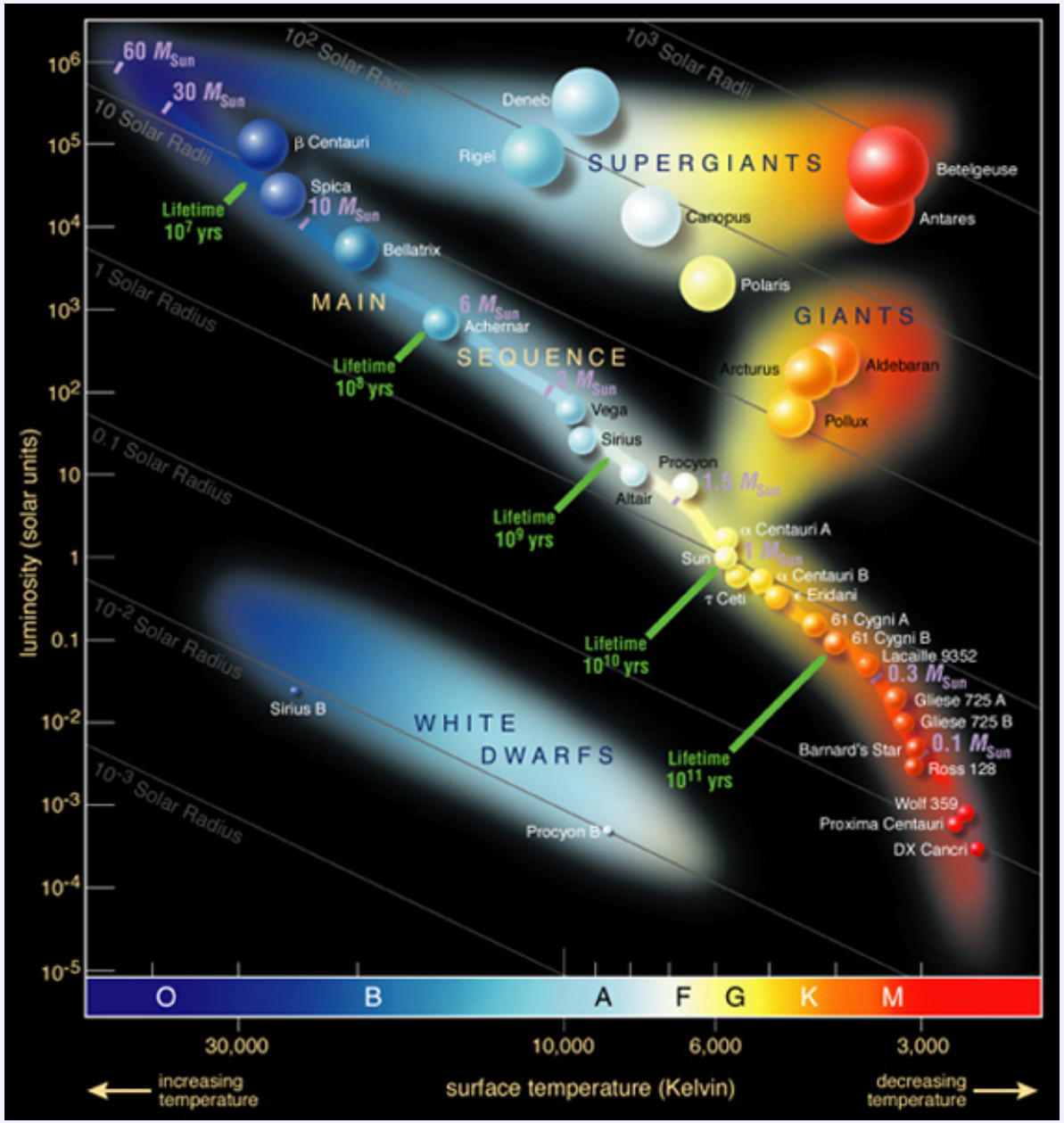

Image credit: ESO

The HR diagram

- Hertzsprung-Russell diagram
- Specific regions:
  - Main sequence
  - Red giants
  - Supergiants
  - White dwarfs
- Links between regions
- Several empty regions
- Luminosity range: $10^{-4} - 10^{5}\,\mathrm{L}_{\odot}$
- Effective temperature range: $2 \cdot 10^{3} - 10^{5}\,\mathrm{K}$
  - Note that the convention is that effective temperature increases to the left
- Radius range: $10^{-2} - 10^{3}\,\mathrm{R}_{\odot}$

Straight lines for fixed radius

- Remember:

$$L = 4\pi\sigma R^2 T_{\mathrm{eff}}^4$$

- Use scaling relation with the Sun:

$$\frac{L}{L_\odot} = \left(\frac{R}{R_\odot}\right)^2 \left(\frac{T_{\mathrm{eff}}}{T_{\mathrm{eff},\odot}}\right)^4$$

- Now take a logarithm:

$$\log\left(\frac{L}{L_\odot}\right) = 2\log\left(\frac{R}{R_\odot}\right) + 4\log\left(\frac{T_{\mathrm{eff}}}{T_{\mathrm{eff},\odot}}\right)$$

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The HR diagram smartphone app

https://astro.unl.edu/smartphone/HRDiagram/

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Life of a star - tracked in the HR diagram

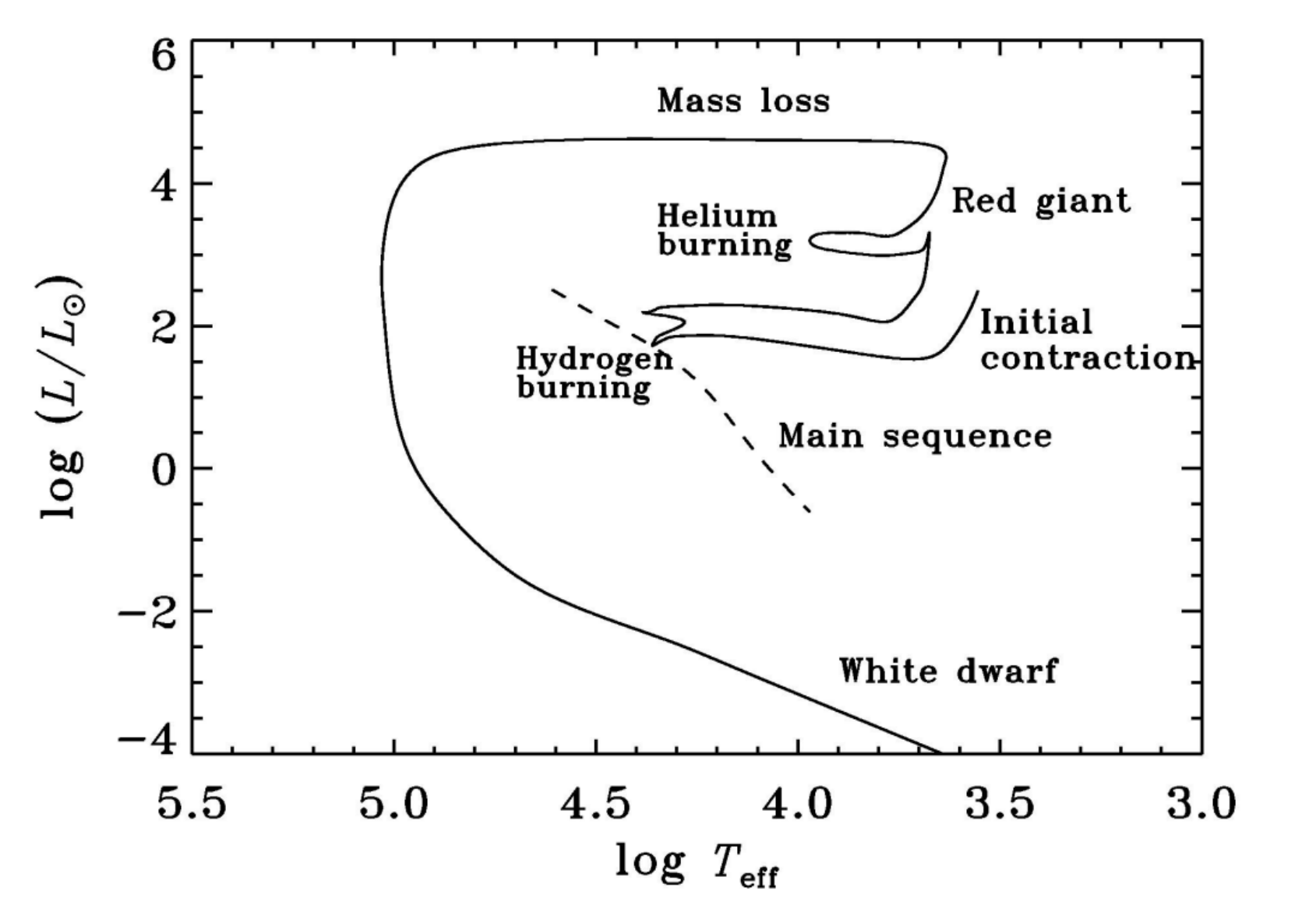

Image credit: JCD

Stellar evolution base concepts

- Constant fight between gravity and nuclear reactions [gravity almost always wins eventually]
- Gravitational contraction releases gravitational potential energy
- Energy is lost [transported] via radiation [or convection]
- Nuclear reactions take place when temperature is high [and density]
- Core contraction leads to outer layer expansion
- Life of a star [stellar evolution] is a repetitive process
- Death of a star is mass-dependent - slido!

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Nuclear fusion in massive stars

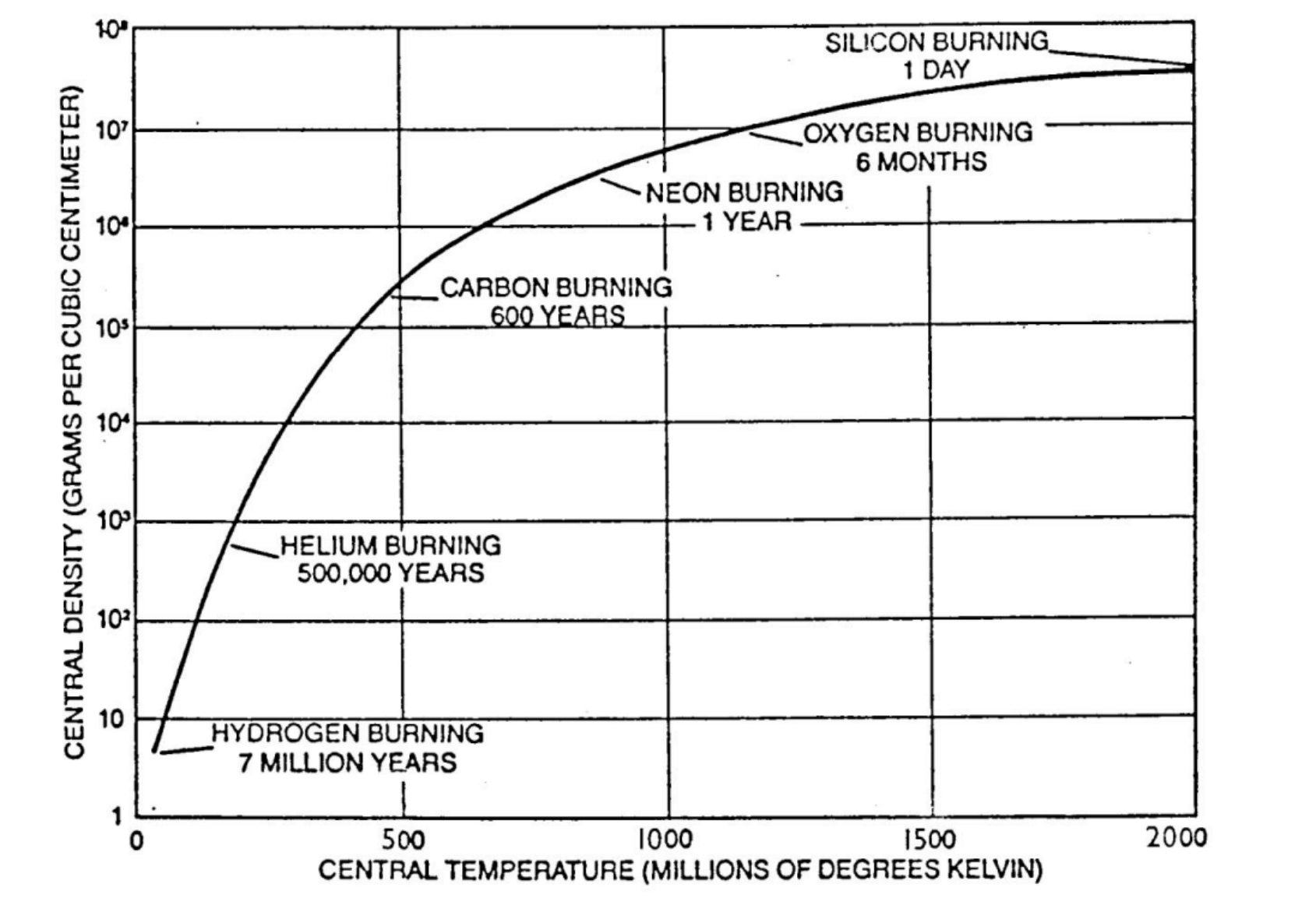

Image credit: JCD

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Stellar modeling assumptions

- Describing stellar structure and evolution highly complex
- Simplifications necessary to describe the physics and do the computations
- Main assumptions:
  - Spherical symmetry
  - Hydrostatic equilibrium
  - Thermodynamic equilibrium
  - Only convection transports chemical elements
  - Ignore: rotation, magnetic field, layer mixing, mass-loss

Combining physics to describe stars

- Thermodynamics - microscopic properties of stellar matter, equation of state
- Hydrodynamics
- Radiation theory - energy transport
- Atomic physics
- Nuclear physics

Describing stellar interiors is complex
But the basics can be described using "simple" physics

Stellar modeling tools - eg MESA

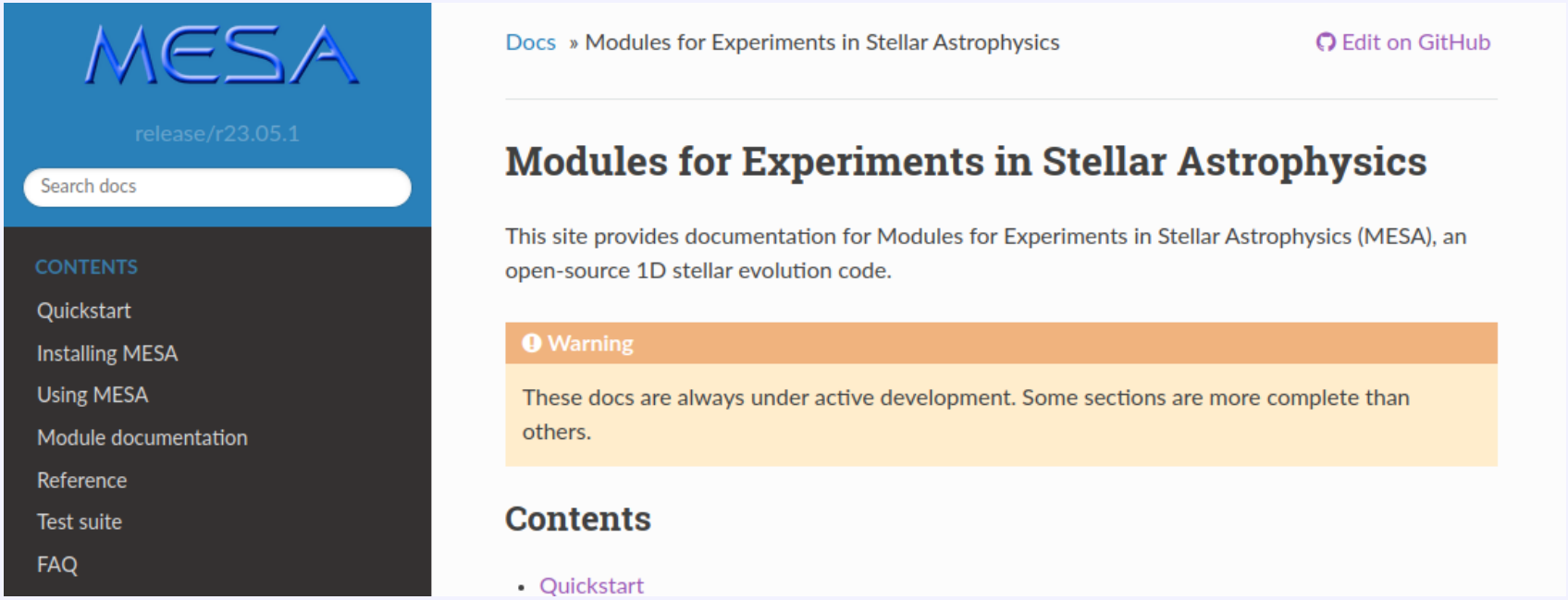
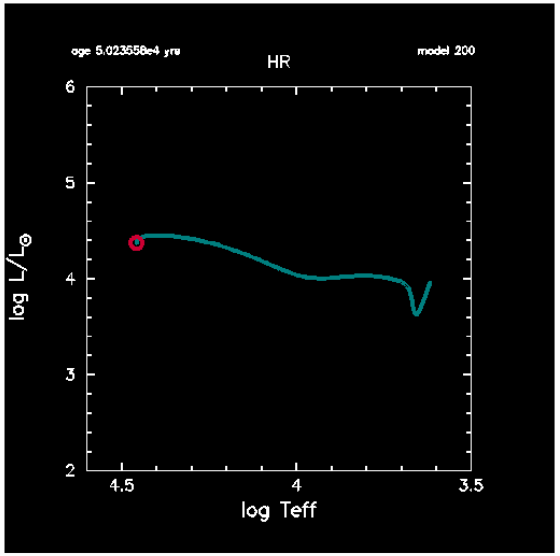


Image credit: MESA

Stellar modeling tools - eg MESA

MESA will also display some pgstar plots that look like:



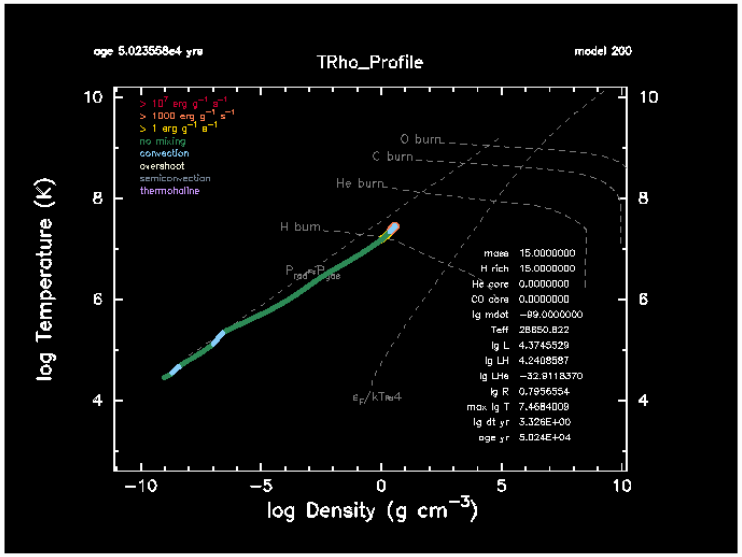


Image credit: MESA

Testing theoretical stellar models

- Usually by testing groups of stars - clusters
- Statistical population studies
- Stars pulsate allowing a "look" inside the star: Asteroseismology

Summary

- Hertzsprung-Russell diagram shows evolution of a star
- Effective temperature of a star defined through its luminosity and radius:

$$L = 4\pi\sigma R^2 T_{\text{eff}}^4$$

- Throughout their life stars move through the HR diagram
- Stellar evolution is mass dependent
- Stellar modeling relies on many simplifications

NEXT LECTURE: Observable properties, Colour-magnitude diagram